

Generative AI Foundations for Engineers

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Module 3: Agents and Tool Use

Agents

- “An agent in generative AI is a system that uses a language model to make decisions, take actions, and interact with tools or environments based on a defined goal or task.”
- An LLM on its own can only use language to do things
- Agents extend this by giving the LLM **tools** for taking action
- Can plan and reason to decide on what tool to use
- Useful for multi-step tasks, orchestrating actions

Types of Agents

- Reasoning and Acting (ReAct)
 - Uses reasoning (thoughts) and actions (tools use) in sequence
 - Useful for explainability and debugging
- Tool-using agents
 - Dynamically choose and call external tools or APIs based on user input
 - Useful for workflows that involve things like calculations, real-time data retrieval
- Chatbots
 - Focus on natural, back-and-forth interactions
 - Useful when personalization and/or memory are required
- Multi-agent systems
 - Involves multiple agents with specialized roles that collaborate with each other and hand off tasks to complete complex objectives

Tools

- The true power behind agentic GenAI!
- External functions or APIs that LLMs can call to perform specific tasks
 - Ex: looking up information, doing math, querying a database, retrieving files from storage
- Extend LLM's capabilities beyond text generation, enabling it to interact with real-world systems, retrieve fresh data, etc.
- Tool descriptions should very clearly state what the tool does, what information it requires, etc.

LangChain and Agents

- Multiple ways to create them and add tools
- LangChain was designed for single agents applications with a handful of tools
- As the number of tools increases, consider going to multi-agent so the agent doesn't get confused which tools to call
- **LangGraph**

GenAI Evaluation

- Many ways to do this
- Ideally have ground truth question-answer pairs
 - The wording of these is very important
- Ideally have many questions (10s - 100s) for full evaluation
 - Automation is important as QA evaluation set grows
- If you are implementing RAG, be sure to have QA pairs specifically adapted to that dataset
- LLMs for evaluation!
 - Tools exist for automating this with built-in evaluators (ex: LangSmith)
 - Can start with very simple evaluators